

Question 32 (a)

Criteria	Marks
• Identifies features that facilitate the transmission of rabies between hosts	2
• Provides some relevant information	1

Sample answer:

The virus is able to travel via the nervous system to the salivary glands. This can result in direct contact transmission when the infected host bites another animal.

Question 32 (b) (i)

Criteria	Marks
• Explains a feature that distinguishes the rabies virus from cellular pathogens	3
• Outlines feature(s) that distinguish between viral and cellular pathogens	2
• Provides some relevant information	1

Sample answer:

The rabies virus has a small genome composed of single-stranded RNA but the genome of cellular pathogens such as bacteria is much larger and is in the form of DNA which enables complex cellular processes without a host.

Question 32 (b) (ii)

Criteria	Marks
• Explains the role of RNA polymerase in reproduction of the rabies virus, including reference to transcription, production of proteins and viral RNA replication • Relates newly produced proteins to RNA polymerase and viral RNA replication	5
• Describes the role of RNA polymerase in reproduction of the rabies virus, including reference to transcription, production of proteins and viral RNA replication • Relates newly produced proteins to RNA polymerase	4
• Outlines the role of RNA polymerase in reproduction of the rabies virus including reference to the processes of transcription and RNA replication	3
• Provides some features of the role of RNA polymerase in reproduction of the rabies virus	2
• Provides some relevant information	1

Sample answer:

The viral RNA polymerase, which is made up of the L and P proteins, is responsible for the production of viral proteins and viral RNA, which are the components of new rabies viral particles.

RNA polymerase is responsible for transcription of the viral RNA into complementary mRNA strands, which are then translated into rabies viral proteins, G, M, N, P and L, by host cell ribosomes.

The replication of rabies viral RNA is facilitated by the newly produced L and P proteins (RNA polymerase). In this process, a complementary strand of viral RNA is produced from the original viral RNA strand. This complementary strand is then used as a template for the RNA polymerase to catalyse the production of more viral RNA. The new strands produced are therefore the same as the original viral RNA.

RNA polymerase is responsible for both the viral RNA and proteins needed in reproducing the virus.

Question 32 (c)

Criteria	Marks
- Accurately interprets and uses the relevant data and information to explain how PEP prevents rabies developing after exposure to the virus - Demonstrates an extensive knowledge of vaccination, passive and active immunity - Explains the role of antibodies - Communicates scientific information succinctly and logically using precise scientific terminology	8
- Interprets and uses the data to explain how PEP prevents rabies developing after exposure to the virus - Demonstrates a thorough knowledge of vaccination, passive and active immunity - Describes the role of antibodies - Communicates scientific information logically using correct terminology	7
- Interprets and uses data provided to explain how PEP prevents rabies - Demonstrates a sound knowledge of vaccination and immunity - Outlines the role of antibodies - Communicates using scientific terms	5–6
- Makes reference to information provided - Provides some information about the role of antibodies and/or vaccines in preventing rabies	3–4
Provides relevant information about immunity and/or rabies	1–2

Sample answer:

Once the rabies virus has entered the wound it will use the patient's cells to replicate and the viral concentration increases (as seen in the first five days). Without PEP the virus will continue to replicate, migrate to the CNS (in the first graph this occurs by day 7), and eventually cause rabies and death.

Initially the patient does not have memory cells for the virus and does not produce antibodies to it. The injection of HRIG provides the patient with human antibodies that are specific to the virus, providing passive immunity. The antibodies will combine with viral molecules and inactivate them and also facilitate their destruction by white cells such as phagocytes. The PEP graph shows these antibodies will only last for up to 21 days but have an important initial role in controlling viral numbers (the data shows a reduction in viral concentration by days 6–8). The antibodies help to prevent the virus progressing to the CNS.

The rabies vaccine contains an inactivated, harmless version of the rabies virus and will result in the development of acquired, active immunity. This occurs when macrophages present the viral antigen to specific helper T cells that have surface receptors complementary to it. These helper T cells then stimulate specific B cells that have also been exposed to the antigen to undergo mitosis and cell differentiation. The daughter cells will either be plasma cells that produce antibodies or will be kept as memory B cells in case of future infection.

The PEP graph shows that by day 7 the production of antibodies begins and increases sharply over the next few days. This coincides with the rapid decrease in viral concentration. The viral particles are being inactivated by the presence of antibodies in increasing concentration.

By day 11 there are no viral particles left at the infection site and progression to the CNS and the disease has been avoided. Antibody concentration continues to increase as more rabies specific plasma cells are produced, with a maximum concentration being reached at 21 days after initial exposure and vaccination.